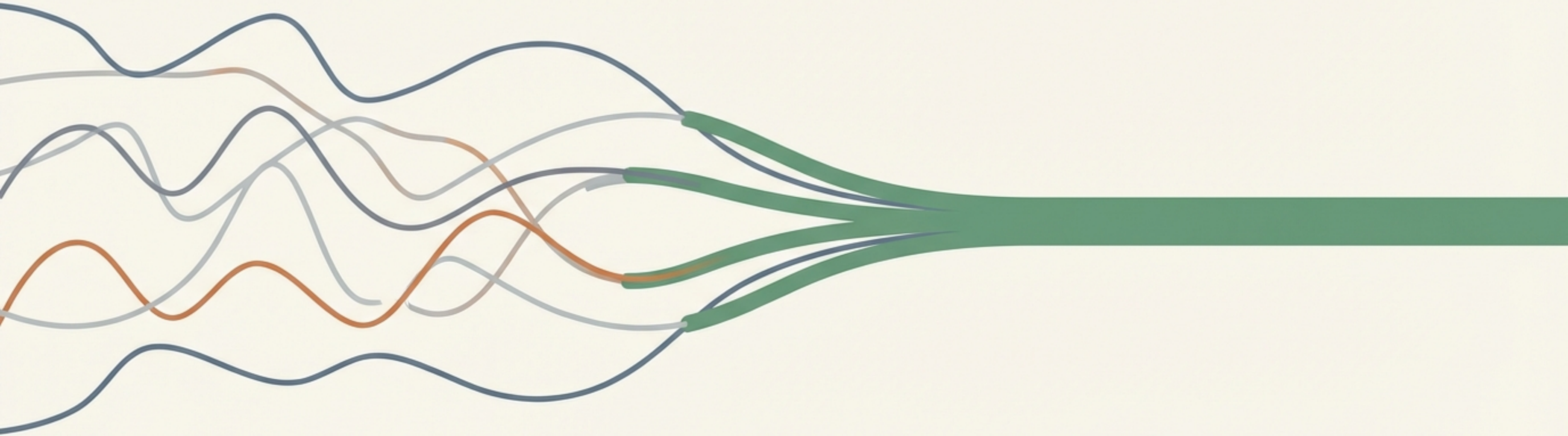


Masterclass in Bagging

Building robust Machine Learning predictions through Bootstrap Aggregating.



Deconstructing the Etymology

[Bootstrap] + [Aggregating] = Bagging

Bootstrap: Sampling subsets of data randomly from a primary dataset.

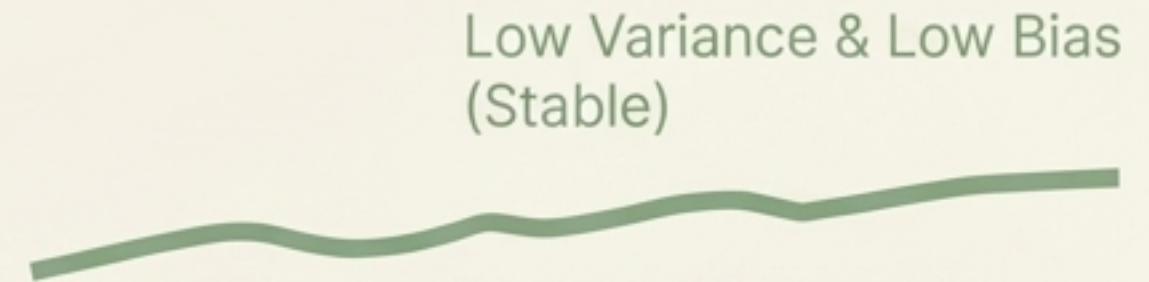
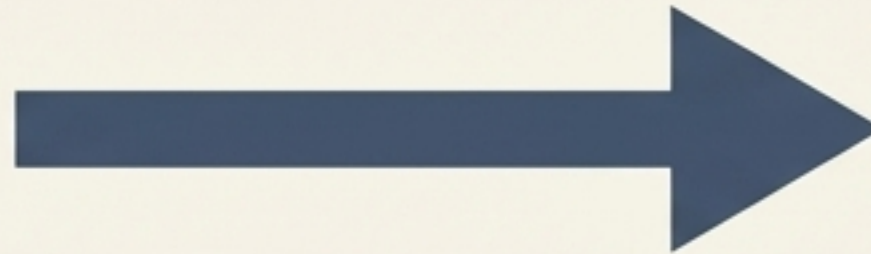
Aggregating: Combining the predictions of multiple models to form a final, singular output.

The Bias-Variance Dilemma

- **The Goal:** Machine Learning requires Low Bias and Low Variance.
- **The Reality:** They are inherently correlated. Decreasing one increases the other.
- **The Culprits:** Complex algorithms (like unpruned Decision Trees) perfectly memorise training data (Low Bias) but fail to generalise to new data (High Variance).



Achieving the Impossible Combination



The Strategy

Keep the algorithm exactly as it is (maintaining Low Bias).

The Intervention

Force the algorithm to produce consistent results, regardless of how the underlying data changes (reducing High Variance).

The Result

A final model boasting both Low Bias and Low Variance.

Step 1: Bootstrapping the Data

Random Sampling

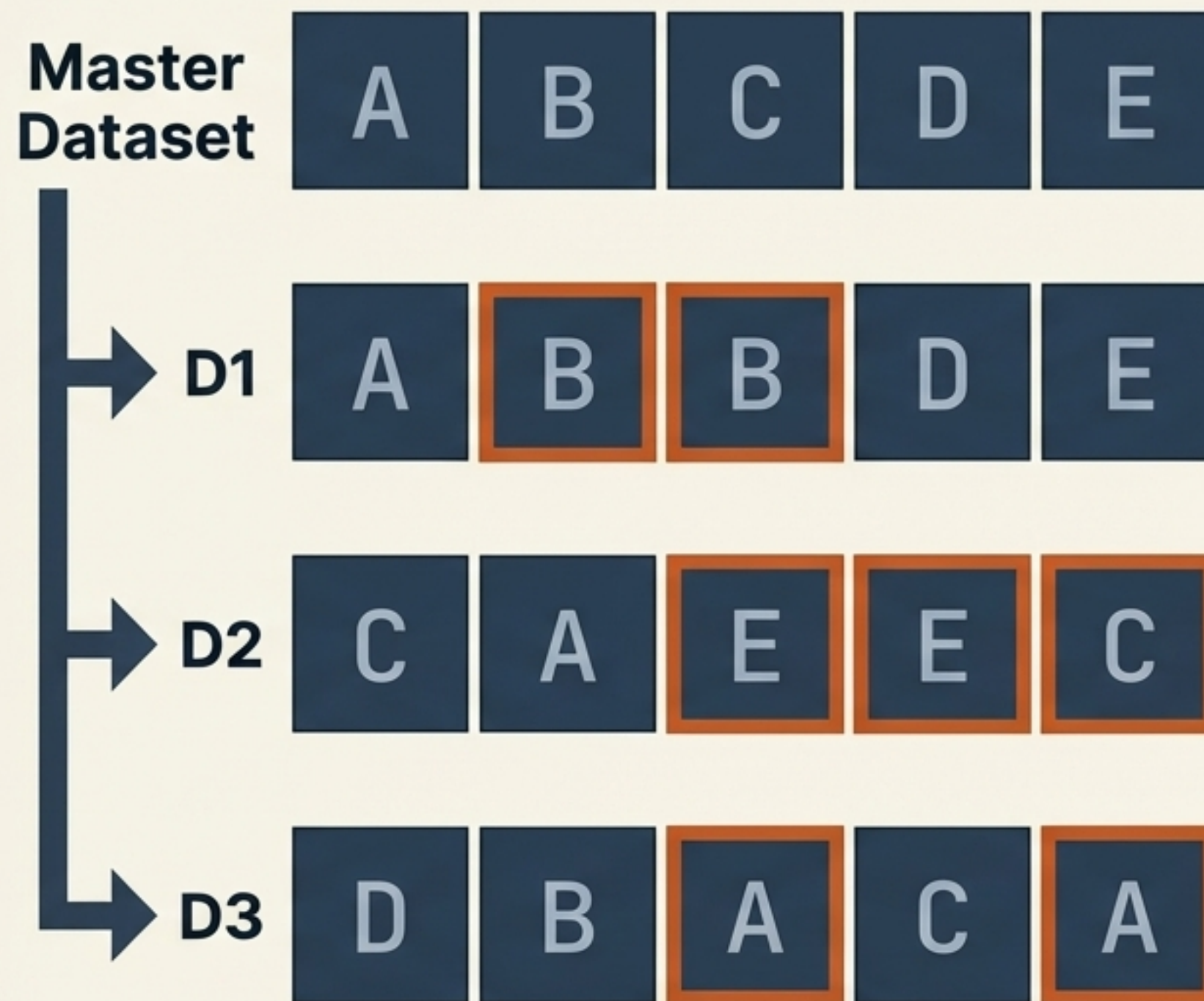
Extract subsets of data from the master dataset.

With Replacement

A single data row can appear multiple times within the same subset.

The Effect

Every new subset is structurally unique, containing variations of repetition and omission.



Step 2: Training Independent Base Models

Uniform Algorithms

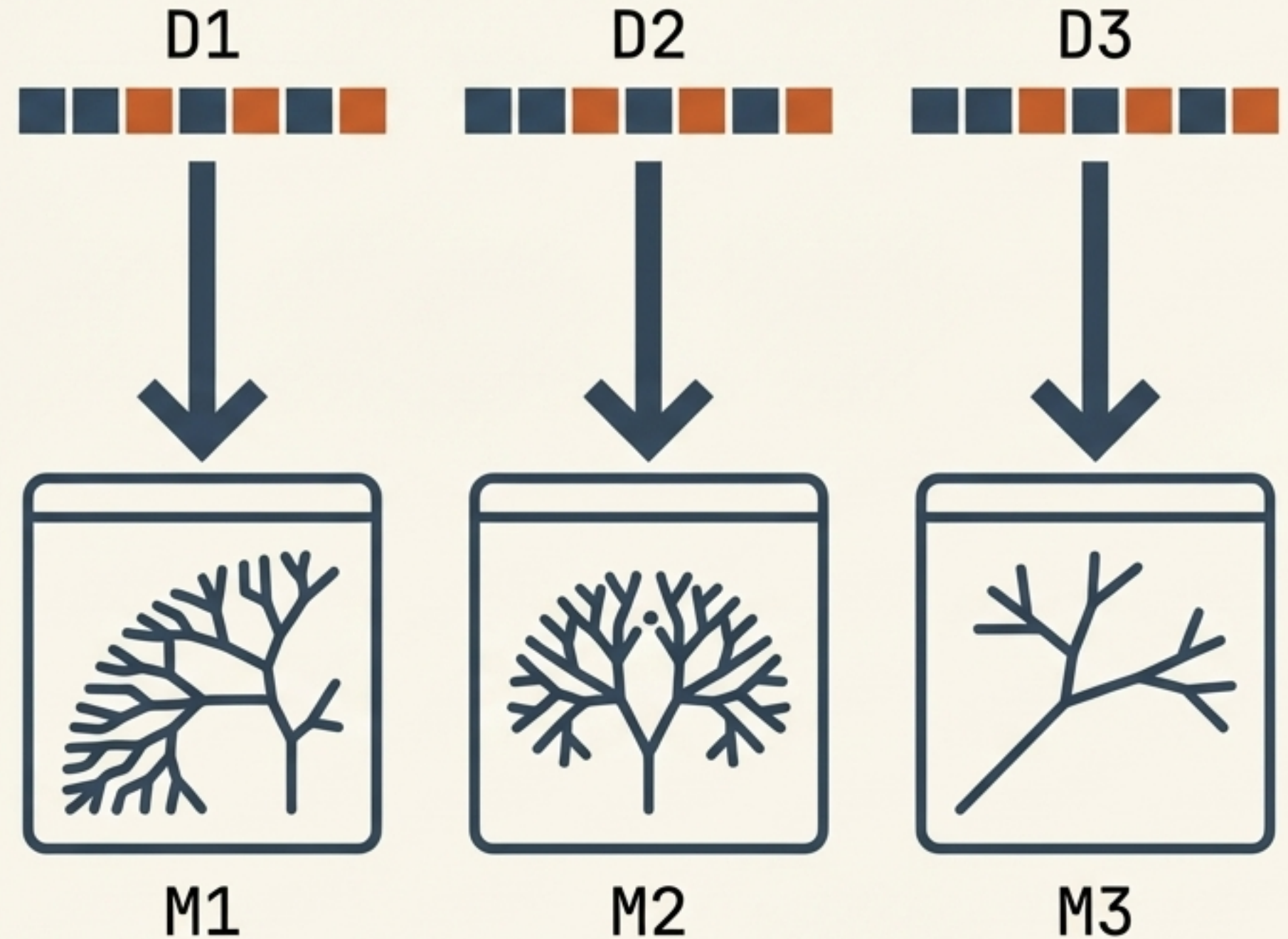
Every base model uses the exact same algorithm (e.g., all Decision Trees).

Diverse Perspectives

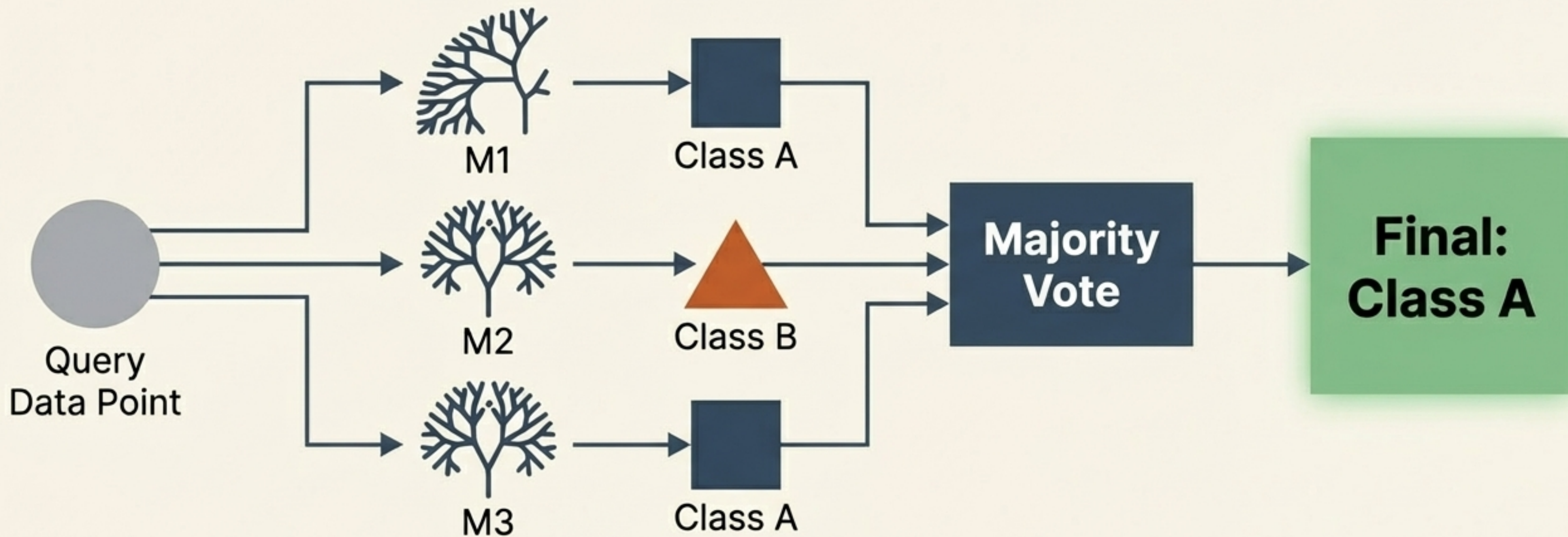
Because they are fed different bootstrapped data (D1, D2, D3), they develop unique predictive rules.

Rule of Thumb

Do not mix algorithms (like blending an SVM with a Tree). Keep the base uniform.



Step 3: Aggregating the Wisdom of the Crowd



The Test

A new, unseen data point is passed through all trained base models simultaneously.

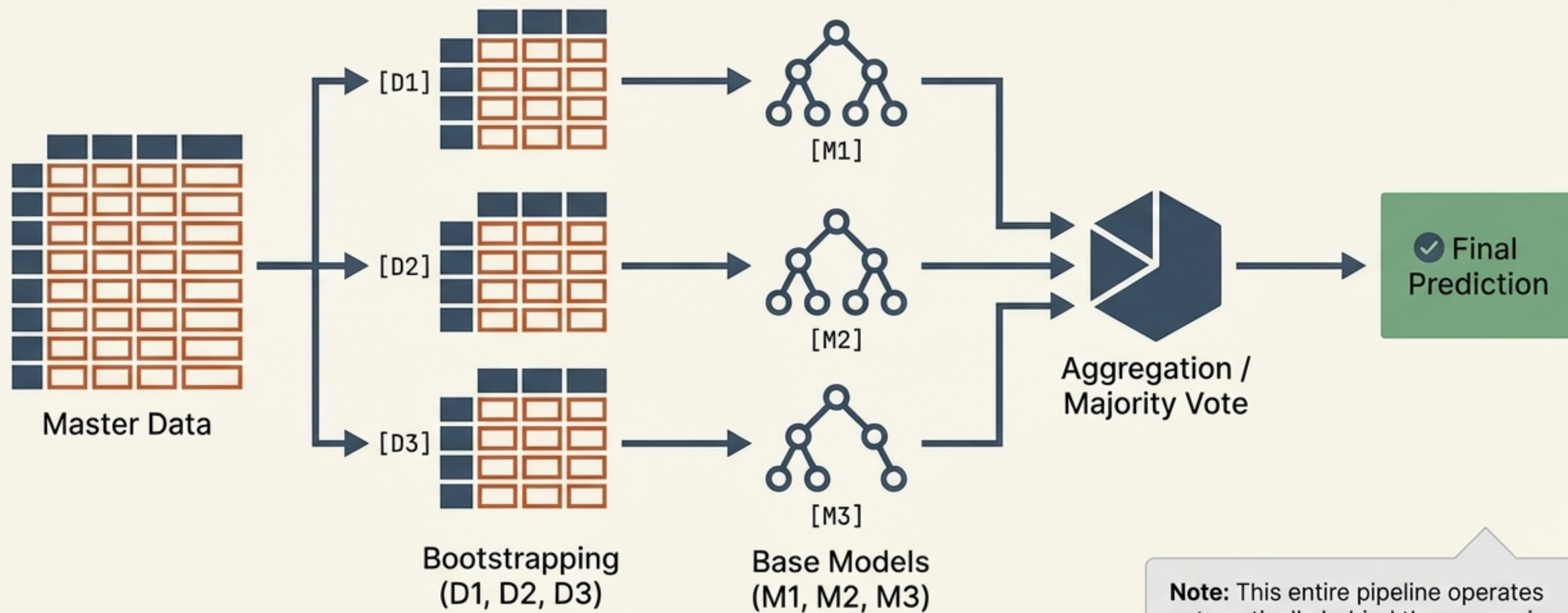
The Decision

For classification tasks, the final output is determined by the Mode (Majority Count).

The Philosophy

The collective majority overrides individual model errors.

The Complete Bagging Architecture



Why Variance Drops in Practice

Diluting the Noise

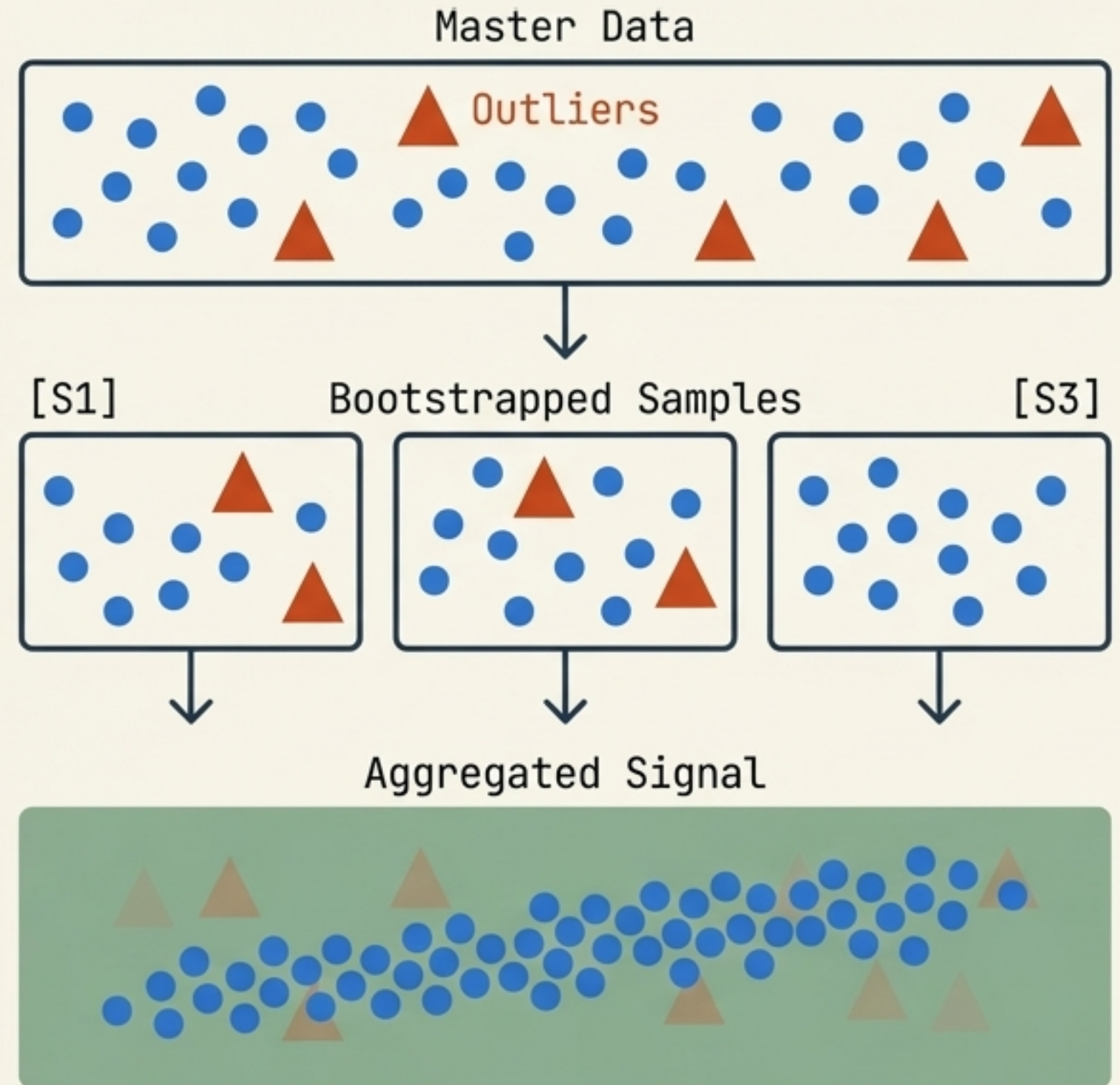
If a dataset has extreme outliers, a single model will overfit to all of them.

Distributing the Error

Bootstrapping scatters these outliers across different base models.

Cancellation

When aggregated, the core data patterns align and stack up, while the scattered, varied errors effectively cancel each other out.



In Practice: Setup and Shuffling

- Dataset

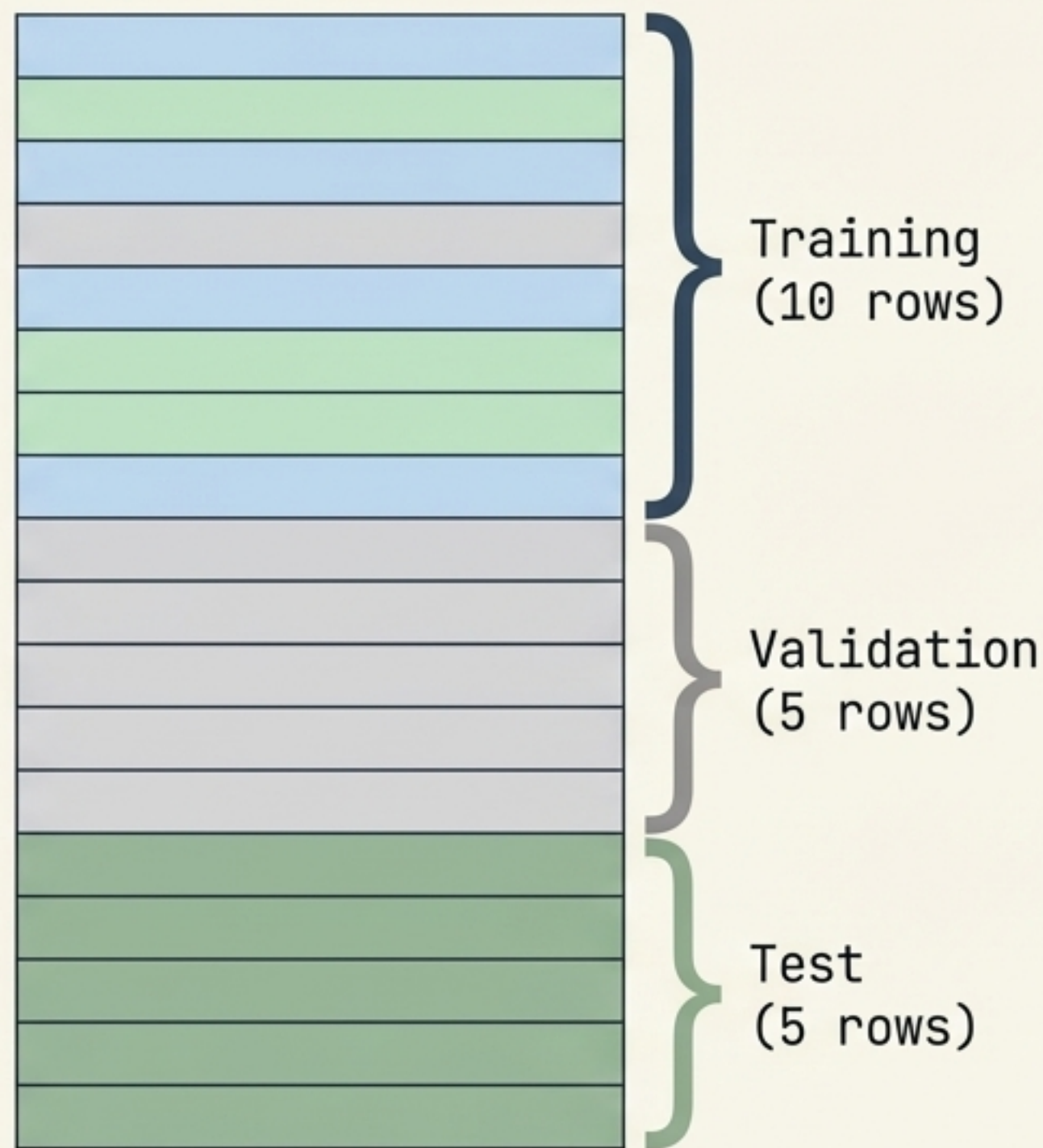
The Iris Classification Dataset.

- Crucial Prep

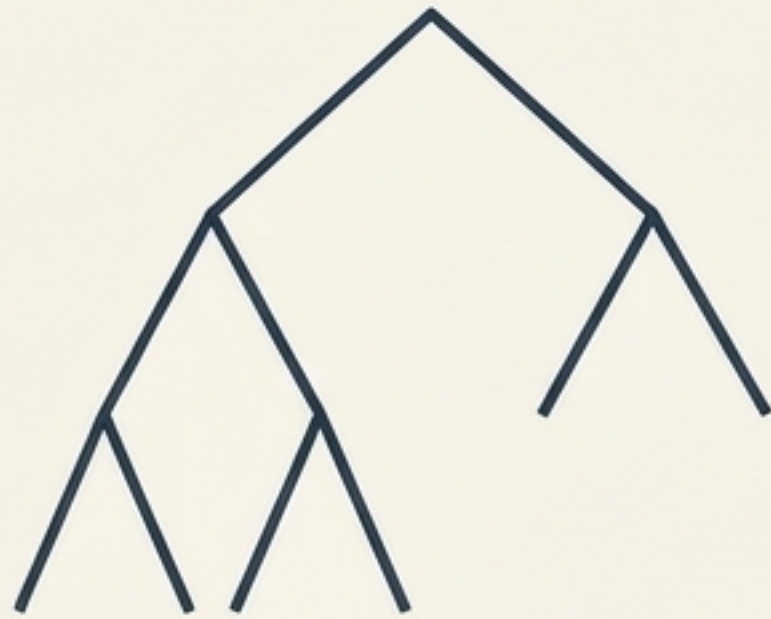
The raw data is inherently ordered by class. It must be thoroughly shuffled to ensure random distribution.

- Demo Parameters

We allocate just 10 random rows for training to clearly observe the bootstrapping effect.

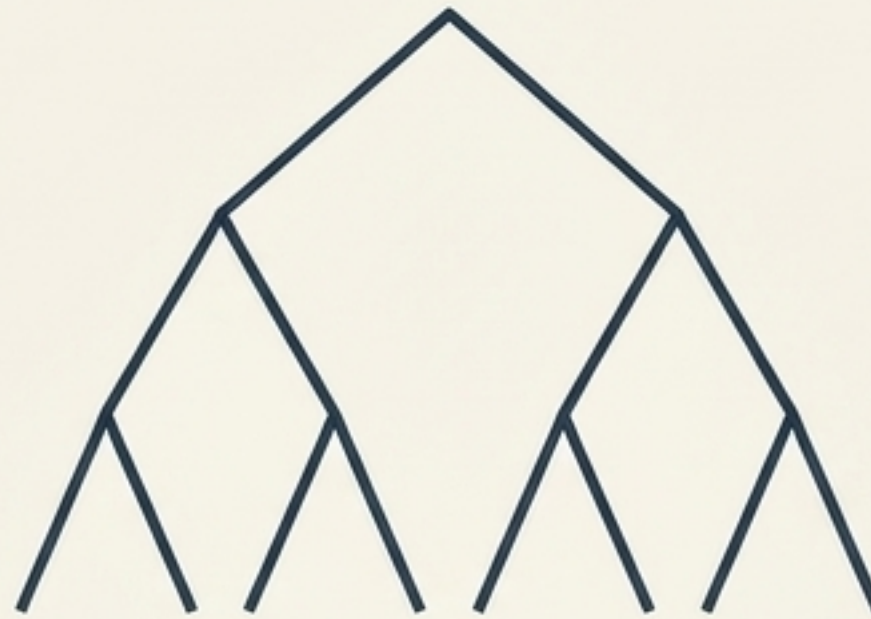


In Practice: Forcing Structural Diversity



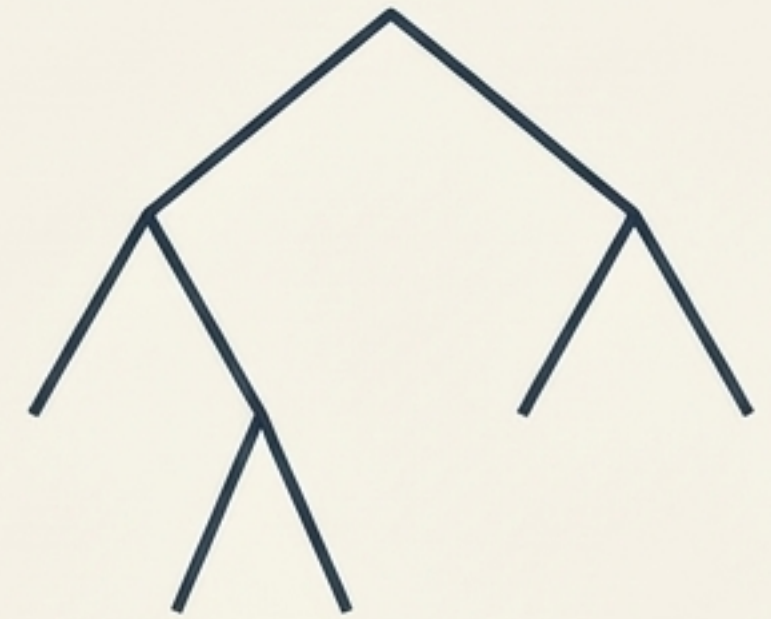
Tree 1

[**Row 92**, **Row 92**, Row 14,
Row 06, Row 22...]



Tree 2

[Row 87, Row 37, Row 47,
Row 11, Row 50...]

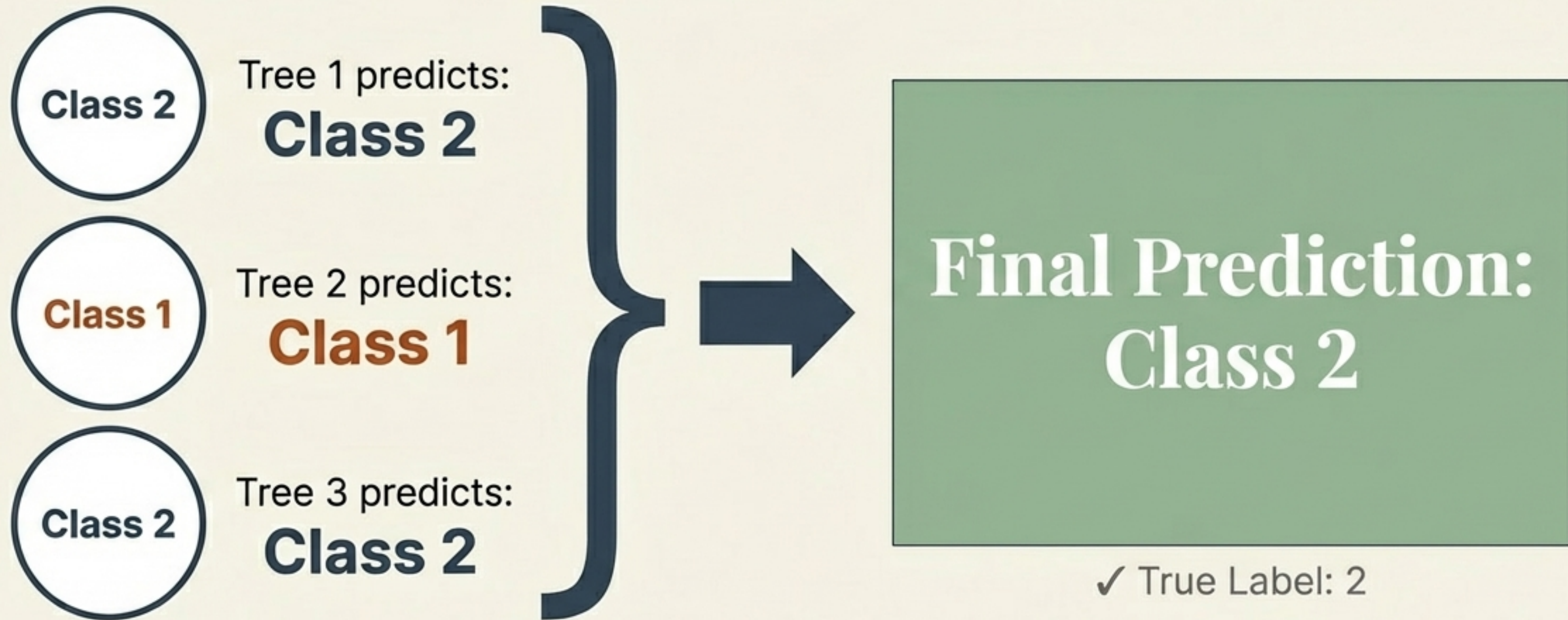


Tree 3

[**Row 97**, **Row 97**, **Row 97**,
Row 03, Row 41...]

Because we sample with replacement, identical base algorithms are forced to build distinct internal decision rules based on heavily varied data exposure.

In Practice: The Final Vote Resolves the Error



Even when one model fundamentally misunderstands the query point (Tree 2), the aggregated consensus absorbs the error and delivers the correct classification.

Creating Variation Beyond the Rows

Traditional Approach

Sampling rows creates variation in data points.

Advanced Approach

For complex datasets (like images or text analysis), sampling columns creates variation in features.

The Four Flavours of Sampling

Data Sample Methods	Bagging Row Sampling With Replacement. (Rows repeat).	Pasting Row Sampling Without Replacement. (Every row is unique).
	Random Subspaces Column Sampling. (Varying the features).	Random Patches Row + Column Sampling. (Varying both data and features).
Feature Sample Methods		

Strategic Deployment



When to use:

Whenever dealing with Low Bias, High Variance algorithms.

The Golden Rule:

Use Bagging to force consistency out of wildly erratic, over-fitted models.

Industry Standard:

The famous 'Random Forest' algorithm is simply a highly optimised implementation of Bagging over Decision Trees.